The Five-Gate Deployment Model

A Deployment Discipline for AI Systems

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Abstract

An institution can be asked two questions about an AI system in production. The first is whether it works. The second is how it got there, and who said it could. The second question is the one a supervisor, an auditor or a board

asks after something has gone wrong, and it is the harder of the two to answer afterwards, because the machinery that put the system into production was built to move work forward rather than to record who accepted what.

This specification defines the **Five-Gate Deployment Model™**, a deployment discipline for AI systems. No AI system reaches production, or remains there, except by passing five gates in order: **G1 Data and Design**, **G2 Validation**, **G3 Approval**, **G4 Deployment** and **G5 Operation**. Each gate carries entry criteria supplied by a named framework, exit criteria, a defined rejection path, one accountable person, and a required evidence record.

Two properties distinguish the model from the machinery it sits beside. **The accountable role at every gate is a single named person and MUST NOT be a committee**, because accountability distributed across a body is accountability no member carries alone, and the question asked after a failure is which person is answerable. And **every gate passage record is an instance of the BOE Declaration** [2]: the entry criteria are the boundary the gate fixes, the deployment freedom granted on passage is the optimizer it frees, and the record is the evidence that the boundary held. That mapping is what allows a gate record to be joined with a validation record, a residency attestation and a vendor screening record during an incident reconstruction, rather than merely filed alongside them.

Two paths run backwards through the model and are part of it rather than exceptions to it. **Rollback** reverses G5 to G4 with the evidence of the reversal preserved. **Loop-back** reopens G2 when an incident occurs at G5, so that a system which failed in operation re-enters validation rather than being patched in place.

The specification states what would falsify the model: that institutions operating five gates with a named accountable person at each deploy unfit AI systems at the same rate as institutions operating an equivalent automated pipeline with no named approver. **No institution unconnected to the author has been observed operating the full sequence**, no readiness instrument exists, and no gate evidence templates are published. Section 8 states each of those plainly.

It is a specification, not a certification scheme, and no conformity assessment body operates against it.

Keywords: AI governance; deployment governance; model risk management; release management; accountability; evidence architecture; gated deployment; regulated industries; MLOps governance.

Notation

This specification uses **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** as the normative verbal forms.

- **MUST** indicates a requirement. A deployment that does not meet it is not conformant with this specification.
- **SHOULD** indicates a recommendation. Departure requires a recorded rationale.
- **MAY** indicates a permission.

The MESA Framework specification [1] and A Pattern Language for Production LLM Platforms [2] use the verbal forms of ISO/IEC Directives Part 2, in which **shall** carries the meaning assigned to MUST here. The correspondence is exact.

AI system. A model, an agent, or a service whose behaviour is materially determined by a model, together with the configuration and policy that bound it. The unit that passes a gate is the AI system, not the model alone, because a model with different guardrails is a different system.

Gate. A checkpoint at which a single named person accepts or rejects the passage of an AI system from one lifecycle stage to the next, on stated entry criteria, and writes a record of the decision.

Gate passage record. The evidence artifact a gate produces: who accepted, on what evidence, when, and under which policy version.

Institution. The accountable legal entity operating the AI system. **Regulated institution** means one whose ability to operate is conditioned on its ability to reconstruct and defend a decision to an external authority.

Sections 2 through 6 are normative. Sections 1, 7, 8 and 9 are informative, except where they state a requirement using **MUST**.

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1. Introduction

1.1 The problem

Every institution that deploys software already has a release process, and most that deploy AI systems run them through it. The process is usually competent at what it was built for: it checks that tests pass, that a change is reviewed, that a rollback exists, and that the change is recorded in a ticket.

It answers the wrong question. A release process establishes that a change was **made correctly**. A supervisor asking about an AI system wants to know that the system was **accepted deliberately**, by someone with the authority to accept it, on evidence that existed at the time. Those are different claims and only the second survives an incident. A ticket showing that a pipeline went green tells an auditor that the pipeline went green.

The gap is not usually rigour. It is that the artifacts a governance question needs are produced by frameworks the release process does not consume: a data classification, an independent validation, a vendor clearance, an incident trigger. Each exists somewhere in the institution. None of them is a precondition of anything.

1.2 What a gate is for

A gate exists to convert an artifact into a decision. Before a gate, a validation report is a document. At a gate, it is either sufficient for a named person to accept the system or it is not, and the answer is recorded.

That conversion is the whole of the model's contribution. **A gate produces no primary evidence of its own.** It attests that other frameworks' evidence was present and was relied on by a person at a moment in time. This is why the model is a spine rather than a discipline: a spine carries load rather than generating it.

1.3 Contribution

1. **Five gates in a normative sequence**, each naming the framework that supplies its entry evidence, so that the model composes with the governance family rather than duplicating it. Section 3.
2. **A single accountable person per gate, stated as a MUST**, with the reasoning for the prohibition on committees stated once rather than five times. Section 2.3.
3. **Two reverse paths specified as part of the model**, rollback and loop-back, rather than as exceptions to it. Section 4.
4. **A mapping from every gate record to the BOE Declaration** [2], which is what makes a gate record joinable with the evidence of frameworks specified elsewhere. Section 5.

1.4 Scope

The model governs the passage of an AI system from design into production and through to retirement, under institutional accountability and evidence requirements. It is calibrated to regulated institutions, and 8.2 states what changes for an institution that is not one.

1.5 Non-goals

- **Not a delivery pipeline.** The model says who accepts and on what. It does not specify build, test, packaging or deployment mechanics, and an institution needs a pipeline as well as this.
- **Not a specification of the frameworks that supply its entry criteria.** What a validation must contain, what a classification must cover and what a monitoring threshold should be belong to the frameworks named at Section 3 and catalogued at [3]. A gate is only as strong as the evidence handed to it.
- **Not a certification scheme.** No conformity assessment body operates against this specification, and no institution can be certified against it.
- **Not a statement of regulatory compliance.** Operating five gates does not establish compliance with any statute or supervisory expectation.
- **Not an empirical claim.** Section 8 states what has not been tested.

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2. Conceptual Model (Normative)

2.1 The sequence is normative

An AI system MUST pass G1, G2, G3, G4 and G5 in that order, and MUST NOT enter a gate whose predecessor it has not passed. A gate passed without its entry evidence is not passed. A deployment that skipped a gate is not a deployment the institution can defend, whatever its technical quality.

The sequence is normative because the gates are not independent checks. Each consumes something the previous one established, and reordering them produces a decision taken on evidence that did not yet exist: an approval before a validation is an approval of an assumption.

2.2 What a gate requires

Every gate MUST carry all six of the following. An institution that operates a checkpoint lacking any of them has a review, which is a useful thing and is not a gate.

#	Element	What it fixes
1	**Entry criteria**	What MUST exist before the gate may be entered, and which framework produces it
2	**Accountable role**	The single named person who accepts or rejects
3	**Decision**	Accept, reject, or accept with recorded conditions
4	**Exit criteria**	What the system is permitted to do once the gate is passed
5	**Evidence record**	The gate passage record, per 2.5
6	**Rejection path**	Where a rejected system goes, and what MUST change before it returns

A gate with no rejection path is not a gate. A checkpoint that can only be passed is a formality, and the institution learns nothing from it. The rejection path is the element easiest to omit, because a gate can be built, staffed and operated for a long time without anyone exercising it, and it is required here for that reason.

2.3 The accountable role is one person

The accountable role at every gate MUST be a single named person. A committee, a function, a team, a board or a role held jointly MUST NOT be recorded as the accountable role.

This is the property that distinguishes the model from a pipeline, and it is stated once here rather than repeated at each gate.

A committee can be consulted. It can review, challenge, advise and decide. It cannot be accountable, because accountability distributed across a body is accountability no member of that body carries alone. The question an authority asks after a failure is not which body approved the system. It is which person is answerable, and an institution that cannot answer that has not assigned accountability; it has recorded a meeting.

Three consequences follow and each is normative.

1. **The person MUST be named in advance**, in the decision-rights assignment of the institution's governance function [3], and not selected at the moment the gate is reached.
2. **A named deputy MUST be part of the appointment**, because a role that cannot be occupied when a system is ready to pass is a role that will be bypassed.
3. **The accountable person at G2 and the accountable person at G3 MUST NOT be the same individual for the same system.** Validation and approval are separated deliberately: the value of an independent validation is destroyed if the person who performed it is also the person who accepts it.

2.4 Entry criteria are evidence, not intent

Entry criteria MUST name artifacts that either exist or do not. Policy, training, intent and scheduled work MUST NOT be recorded as satisfying an entry criterion.

A gate whose criteria can be satisfied by a statement of intention is a gate that will be satisfied by a statement of intention. This rule is inherited rather than invented: it is the same rule the MESA Framework applies through its

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enforcement constraint, under which a control the technical substrate cannot enforce operates at the maturity of the substrate rather than at the maturity of the policy [1].

Where an entry criterion cannot be met and the institution nonetheless intends to proceed, the gate MUST be recorded as **passed with conditions**, the conditions MUST be dated and owned by a named person, and the accountable person for the next gate MUST be shown the open conditions before deciding. **An unrecorded exception is a skipped gate.**

2.5 The gate passage record

Every gate MUST produce one record carrying, at minimum:

- the AI system and the version of it that passed;
- the gate;
- the decision, and any conditions attached to it;
- the name of the accountable person who took it;
- the entry evidence relied on, identified so that it can be retrieved;
- the policy version in force at the time of the decision;
- the date and time.

The policy version is the field easiest to omit, because it is a property of the institution at the time rather than of the decision itself, and it is the one an incident reconstruction most needs. A record showing that a person accepted a system on stated evidence is incomplete without the rules that were in force when they accepted it, because the question after a failure is frequently whether the decision was correct under the policy of the day rather than under the policy of today.

2.6 Naming and scope

This specification defines the **Five-Gate Deployment Model™**, a deployment discipline for AI systems. The model governs the passage of a model, agent or AI-enabled service from design into production and through to retirement, under institutional accountability and evidence requirements.

The model is unrelated to, and not derived from, stage-gate processes for new product development. That term is used in the product-innovation field, is asserted as a registered trademark by its proprietor, and denotes an idea-to-launch process for product innovation: a different subject, in a different field, answering a different question. **No affiliation, endorsement or lineage is claimed or implied.**

The terms *stage-gate*, *phase-gate* and *gating process* MUST NOT be used as synonyms for Five-Gate on any surface, including this specification, the registry entry at [3], derived material and commentary. **First mention in any document is *the Five-Gate Deployment Model™, a deployment discipline for AI systems*; subsequent mentions are *Five-Gate*.**

A note on what this paragraph does not say. It does not name the proprietor of the third-party mark or state where that mark is registered. No trademark register was reachable, at the time this specification was prepared, in a way that could distinguish a genuine record from a deliberately invented control, and an attribution carried in the author's own working documents conflicts with the proprietor's current published notice. A specification whose purpose here is good-faith distinct use is the wrong place to print an ownership claim that cannot be verified, and the distinction the paragraph draws does not depend on one.

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3. The Five Gates (Normative)

3.1 Figure 1: the sequence

3.2 G1 Data and Design

Purpose. To establish that the AI system may be built at all on the data and components proposed: that the data is classified and permitted, that its lineage is provable, and that any third-party component has been assessed.

Entry criteria.

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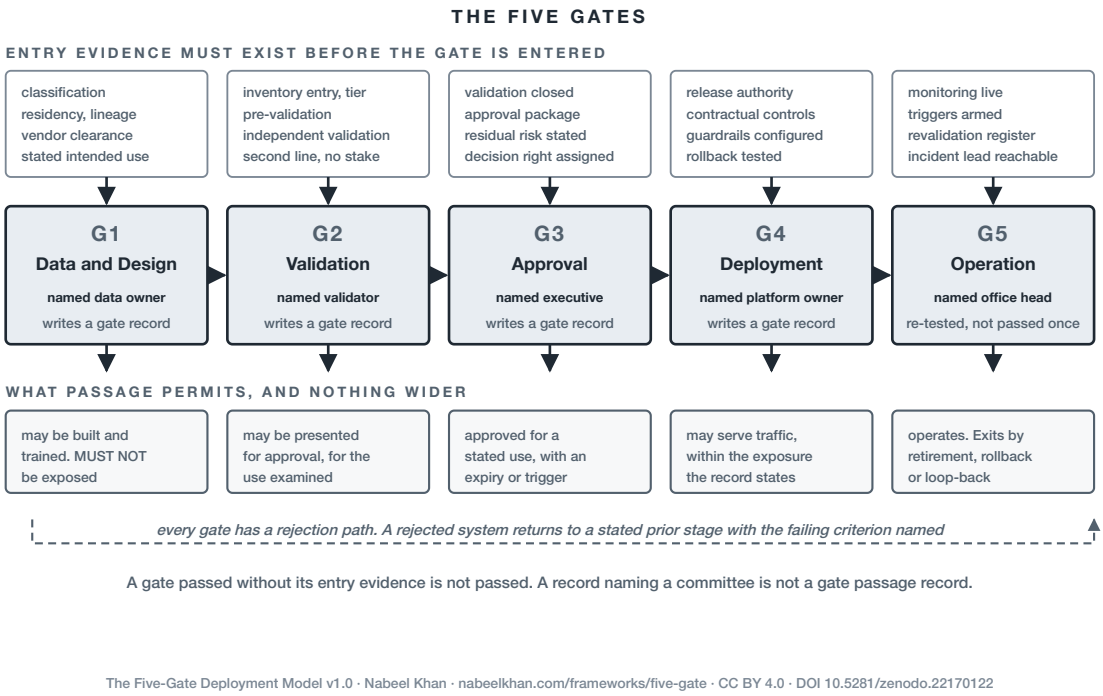


Figure 1. The five gates in sequence. Each gate names the framework supplying its entry evidence, one accountable person, and the record it writes. Rejection returns the system to the preceding stage with a stated reason; rollback reverses G5 to G4 with evidence preserved; loop-back reopens G2 after an incident at G5.

Criterion	Supplied by
Data classification for every data set the system will use	AI Data Governance [3]
Residency determination and attestation per jurisdiction	AI Data Governance [3]
Lineage reaching from an attested source to the point of use	AI Data Governance [3]
Vendor clearance, where any component is third-party	AI Vendor Risk (AVRF) [3]
The deployment topology, where the system spans jurisdictions	Cross-Border AI Architecture Patterns [3]
A stated intended use, and the decisions the system will influence	The institution

Accountable role. The named data owner for the domain the system draws on. Where the system draws on more than one domain, one owner is named as accountable and the others are recorded as consulted; **two accountable owners is the condition under which neither decides.**

Decision. Accept, reject, or accept with conditions.

Exit criteria. On passage, the system MAY be built and trained or configured against the classified data. It MUST NOT be exposed to any consumer, internal or external.

Rejection path. A rejected system returns to design. The rejection record MUST name which criterion failed. A system rejected for an unclassified data set MUST NOT re-enter G1 until the classification exists, and not merely until it is scheduled.

Evidence record. Gate passage record per 2.5, citing each entry artifact.

3.3 G2 Validation

Purpose. To establish that the system performs as claimed, under review by someone with no stake in it passing.

Entry criteria.

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Criterion	Supplied by
The system registered in the model inventory, with an owner and a risk tier	MESA MRM [3]
Pre-validation: data lineage, stated assumptions, stated limits	MESA MRM [3]
An independent validation performed by the second line	MESA MRM [3]
Where a Sharia Supervisory Board holds binding approval power, the Sharia validation track opened	SACF [3]

Accountable role. The named validator. **This person MUST NOT be a member of the team that built the system**, and MUST NOT be the accountable person at G3 for the same system.

Decision. Accept, reject, or accept with conditions. A validation finding that the system is fit for a narrower use than proposed is an acceptance with conditions, and the narrowing MUST be carried into the exit criteria rather than noted in prose.

Exit criteria. On passage, the system MAY be presented for approval. Its permitted use is the use the validation examined, and no wider.

Rejection path. A rejected system returns to design or to G1, depending on whether the failure was in the system or in the data it was built on. The rejection record MUST state which.

Evidence record. Gate passage record, citing the validation report by identifier.

3.4 G3 Approval

Purpose. To establish that a person with institutional authority has accepted the system into production use, on the evidence, and is answerable for that acceptance.

Entry criteria.

Criterion	Supplied by
Independent validation closed	MESA MRM [3]
The approval package: intended use, validated limits, residual risk, open conditions	MESA MRM [3]
Where SACF applies, **both** validation tracks closed, civil and Sharia	SACF [3]
The approval authority itself, assigned in advance	AI Governance Operating Model [3]

Accountable role. The named accountable executive to whom the decision right is assigned. **This is the gate at which the model's central property is exercised**, and 2.3 applies to it without exception.

Decision. Accept, reject, or accept with conditions.

Exit criteria. On passage, the system is approved for the stated use, for a stated period, subject to any conditions recorded. **An approval MUST carry an expiry or a revalidation trigger**, because an approval that never expires is an approval nobody will revisit.

Rejection path. A rejected system returns to G2 with the reason recorded. An approver who rejects for a reason the validation did not examine MUST record that as a finding against the validation scope, so that the next validation examines it.

Evidence record. Gate passage record, naming the person. **A record naming a committee does not satisfy 2.3 and is not a gate passage record.**

3.5 G4 Deployment

Purpose. To establish that the approved system is being placed into production in the configuration that was approved, with its guardrails in force and its exposure bounded.

Entry criteria.

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Criterion	Supplied by
Release authority for the environment	AI Governance Operating Model [3]
Contractual controls in force, where any component is third-party	AVRF [3]
The approved deployment topology, where the system spans jurisdictions	Cross-Border AI Architecture Patterns [3]
Guardrails configured and demonstrated in the target environment	The platform
A tested rollback path	The platform

Accountable role. The named platform owner for the target environment.

Decision. Accept, reject, or accept with conditions.

Exit criteria. On passage, the system MAY serve production traffic, within the exposure the record states. **Staged exposure is the expected form.** Where exposure is staged, each stage MUST carry its own entry condition and the record MUST state what will trigger the next stage and what will trigger a return.

Rejection path. A rejected deployment returns to G3 only if the configuration differs from what was approved. If the configuration matches and the environment is not ready, the system remains at G4 and the record states why.

Evidence record. Gate passage record, citing the approved configuration and the rollback test.

3.6 G5 Operation

Purpose. To establish that the system in production remains within the conditions under which it was approved, and that the institution will know promptly when it does not.

Entry criteria.

Criterion	Supplied by
Monitoring live, covering the conditions the approval was conditioned on	The platform
Incident detection triggers armed	AIRP [3]
The revalidation trigger register populated	MESA MRM [3]
The named incident lead appointed and reachable	AIRP [3]

Accountable role. The named head of the governance office.

Decision. Accept, reject, or accept with conditions. **G5 is the only gate that is not passed once.** It is entered on deployment and is re-tested at every revalidation trigger, and a system that fails G5 leaves production by the rollback path at Section 4.1.

Exit criteria. The system operates. Its exit from G5 is retirement, rollback, or loop-back to G2.

Rejection path. A system that cannot arm its triggers or cannot be monitored against its approval conditions MUST NOT pass G5, and MUST be returned to G4 rather than left running while monitoring is arranged.

Evidence record. Gate passage record on entry, and a dated record at each revalidation trigger.

3.7 What the gates do not do

The gates do not evaluate. **Each gate accepts or rejects on evidence produced elsewhere**, and an institution operating the gates over frameworks that produce no evidence has a sequence of meetings. The strength of a gate is the strength of the framework behind its entry criteria, which is why Section 3 names a supplier for every criterion rather than describing the criterion in the abstract.

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4. Reverse Paths (Normative)

Both paths below are part of the model. An institution that operates the five gates forward and has no defined way back has half a model, and the half it lacks is the half that operates after something goes wrong.

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4.1 Rollback: G5 to G4

A system in operation MUST be capable of being returned to G4, meaning withdrawn from production traffic, by a decision of the accountable person at G5 or the accountable person at G3.

Three requirements attach.

1. **The evidence of the reversal MUST be preserved.** A rollback that removes the system and its records leaves the institution unable to reconstruct what the system did while it was live, which is precisely the period an authority will ask about.
2. **The rollback path MUST have been tested** before G4 was passed, per 3.5. A rollback described and never exercised is a plan rather than a capability.
3. **A rollback is a gate decision and MUST be recorded as one**, with a named person and a stated reason. A withdrawal performed as an operational action and recorded only in a deployment log is not a gate decision.

Rollback does not invalidate the G3 approval. The system remains approved and is not in production, and it returns to production by passing G4 again.

4.2 Loop-back: an incident at G5 reopens G2

An incident that calls the system's fitness into question MUST reopen G2, not merely produce a fix.

This is the model's strongest requirement and the one with the most direct incentive against it, because it is slower than patching. The reasoning is that an incident is evidence about the validation as much as about the system: either the validation examined the failure mode and judged it acceptable, in which case the judgement needs revisiting, or it did not examine it, in which case its scope needs widening. A patch applied without reopening validation changes the system and leaves the assessment of the system untouched, and the institution's belief about the system is then older than the system.

Which incidents trigger loop-back MUST be defined in advance, in the revalidation trigger register at 3.6, and MUST NOT be decided case by case after an incident by the people whose work would be reopened.

4.3 Retirement

A system leaves the model by retirement. **Retirement MUST be a recorded decision by the accountable person at G3 or their successor in the role**, and the evidence records of every gate the system passed MUST be retained for the period the institution's obligations require, which will usually outlast the system.

4.4 Figure 2: the reverse paths

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5. The BOE Mapping (Normative)

5.1 Every gate record is a BOE Declaration

A Pattern Language for Production LLM Platforms [2] names the **BOE Declaration**: a statement of the **B**oundary a control fixes, the **O**ptimizer it frees, and the **E**vidence proving the boundary held. **Every gate in this model MUST state its passage record as an instance of that declaration.**

Gate	Boundary fixed	Optimizer freed	Evidence
G1	Data that is unclassified, unpermitted or unattested may not enter the system	Free choice of architecture, model and method over the permitted data	Classification, residency attestation, lineage, vendor clearance, cited in the record
G2	A system whose behaviour has not been independently examined may not be presented for approval	Free choice of how the system achieves its result within the validated envelope	The validation report, cited by identifier
G3	A system nobody is answerable for may not enter production	Discretion over which systems the institution runs at all	The approval record, naming a person

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Gate	Boundary fixed	Optimizer freed	Evidence
G4	An approved configuration may not be altered on the way to production	Free choice of deployment mechanics, staging and infrastructure	The configuration cited in the record, and the rollback test
G5	A system that cannot be monitored against its approval conditions may not remain in production	Autonomy inside the conditions the approval set	The monitoring record and the armed triggers

5.2 Why the mapping matters

This is not a presentational device. The mapping is what makes a gate record **joinable** with the evidence of frameworks specified elsewhere.

An incident reconstruction under AIRP [3] consumes gate records, validation records, residency attestations and vendor screening records together. It can do so only if all of them state the same three things in the same shape: which constraint was fixed, what was left free, and what shows the constraint held. Two controls that emit evidence in one shape compose. Two that do not are adjacent, however well each is drawn [2].

5.3 The invariant this model instantiates

The Boundary Invariant [2] holds that a boundary is a clause the optimizer may not cross, and everything else is optimization. **A gate is that invariant applied to a moment in time rather than to a mechanism.** The entry criteria are the clause; the freedom granted on passage is what the institution may then optimize; and the record is the proof that the clause held on the day. Read that way, the five gates are five boundaries placed along a lifecycle, and the sequence is the order in which they bind.

5.4 Figure 3: the mapping

---

6. Conformance (Normative)

6.1 Self-declared, and nothing else

Conformance is **self-declared**. No conformity assessment body operates against this specification, no third party is accredited to assess a claim made under it, and no certification exists.

6.2 What a conformance claim requires

An institution conforms, for a stated set of AI systems, when all of the following hold and are evidenced for every system in that set.

1. The system passed G1 to G5 in order, and no gate was entered before its predecessor was passed.
2. Every gate passage record exists and carries every field required by 2.5.
3. Every accountable role was held by a single named person, per 2.3, and G2 and G3 were held by different people.
4. Every entry criterion was satisfied by an artifact that existed at the time of the decision, per 2.4, or was recorded as an open condition owned by a named person.
5. A rollback path was tested before G4 was passed.
6. The revalidation triggers that reopen G2 were defined before G5 was entered.
7. Every gate record states its BOE Declaration, per 5.1.

6.3 What MUST accompany a claim

A published claim MUST state: the set of systems claimed, the systems deliberately excluded and why, the date of the evidence relied on, and the version of this specification claimed against.

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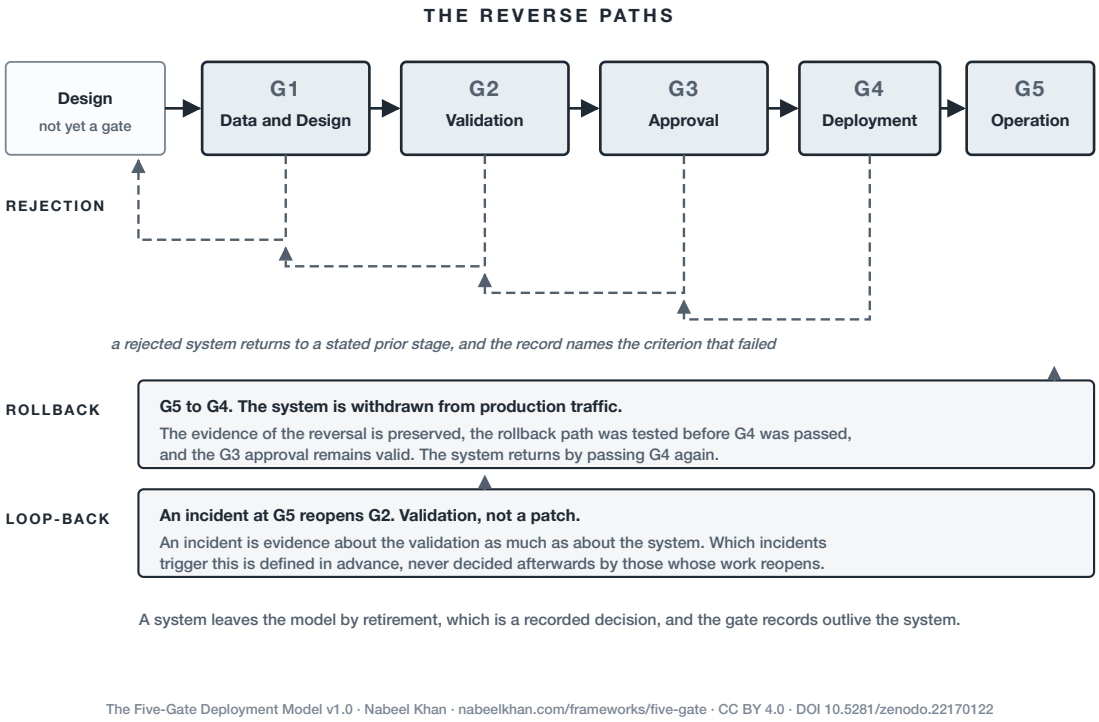


Figure 2. The reverse paths. Rejection at each gate returns the system to a stated prior stage. Rollback withdraws an operating system from production to G4 with its evidence preserved. Loop-back reopens validation at G2 after an incident, so that the institution's assessment of a system is never older than the system.

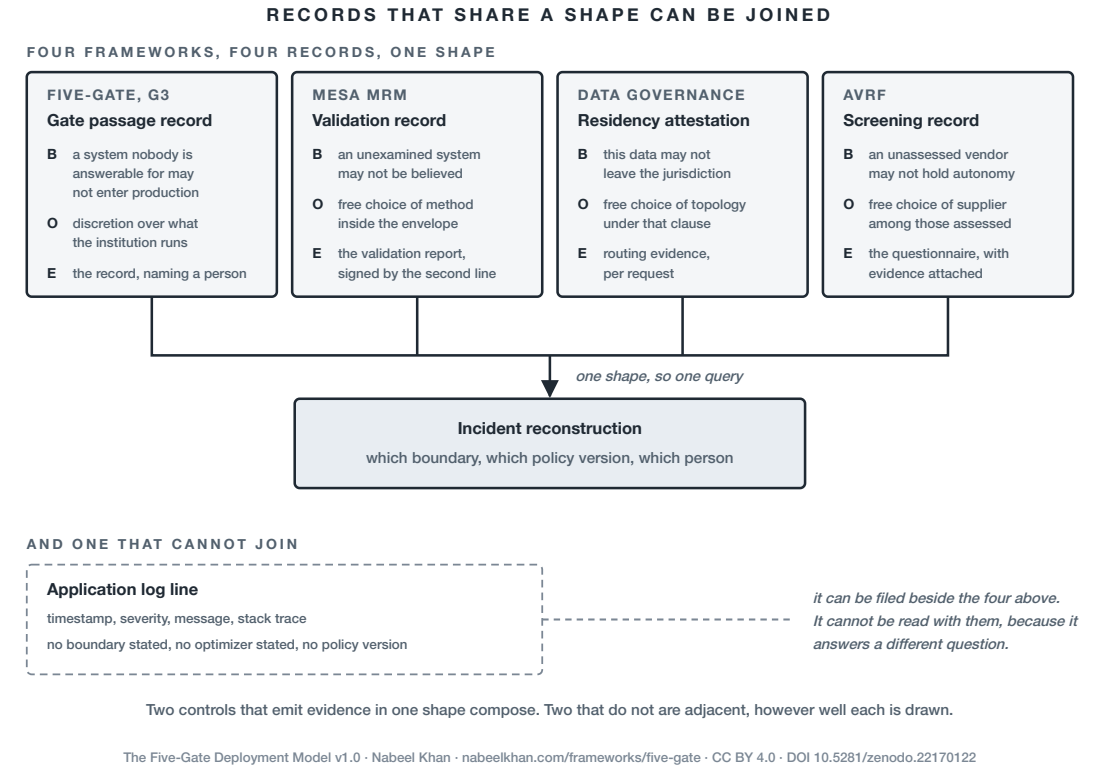


Figure 3. The BOE mapping. Each gate fixes a boundary, frees an optimizer, and writes the evidence that the boundary held. Because all five records state the same three things in the same shape, they join with the records of the frameworks supplying the entry criteria, and an incident reconstruction can read across all of them.

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A claim MUST NOT be made for an institution as a whole. Conformance is a property of a deployment, and an institution-wide claim conceals the systems that did not pass.

6.4 What a conformance claim does not mean

- It does not mean the institution is compliant with any statute or supervisory expectation.
- It does not mean the systems are fit. The gates record acceptance; the frameworks behind them assess fitness.
- It does not mean the institution has been assessed. Nobody assessed it. It declared.
- It does not mean a certification exists. None does.

6.5 Lapse

An institution ceases to be conformant for a system from the date the system's monitoring stopped, its approval expired without revalidation, or a gate condition passed its date unmet, **not from the date the lapse was discovered**.

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7. Related Work

7.1 Model risk management

Supervisory model risk guidance is the closest instrument to G2 and G3, and this model does not duplicate it. In Canada, OSFI Guideline E-23 [7], published 11 September 2025 and effective 1 May 2027, extends model risk expectations to artificial intelligence and machine learning models and to models and data sourced from third parties. In the United States, SR 26-2 [8], issued 17 April 2026, supersedes and replaces SR 11-7 (2011).

The relationship runs one way. An institution subject to either instrument will find that G2 and G3 give its model risk discipline a lifecycle position and an accountable person, and **MUST perform its own crosswalk against the primary source in force**. This specification does not state either instrument's requirements and does not claim to satisfy them.

7.2 Management system standards

ISO/IEC 42001 [5] specifies a management system for artificial intelligence and the NIST AI Risk Management Framework [6] specifies a risk management structure. Both operate at institutional altitude. Neither specifies a deployment sequence with named accountable persons, which is this model's subject. An institution certified to ISO/IEC 42001 may operate this model, or not, and this specification does not change that.

7.3 Delivery pipelines

A continuous delivery pipeline enforces a sequence and can enforce it far more reliably than a human process. **The difference is not rigour and this specification does not claim to be more rigorous than a pipeline.** The difference is that a pipeline records that a condition was met and a gate records that a person accepted it. Where an institution can automate an entry criterion it SHOULD, and the automated check then becomes the artifact the accountable person relies on rather than a substitute for the decision.

The strongest form of this model is a pipeline that enforces the entry criteria and a person who accepts the passage, and 8.3 records that this combination has not been evaluated against either alternative.

7.4 The governance family

The model is entry REG-06 of the Defensible AI Framework Registry [3], which is the canonical source for its relationships to the other frameworks named throughout Section 3. **Where this specification and the registry describe the same relationship, the registry governs**, because two documents asserting the same relationships will otherwise drift.

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8. Limitations and What Has Not Been Tested

8.1 No institution unconnected to the author has been observed operating the full sequence

This is stated first because it is the most important. The model is named and used, and no third party has published a result obtained under it. Everything in Sections 2 to 5 is an architectural claim. It is coherent, which can be checked from the document, and it is not thereby correct.

8.2 It is calibrated to institutions that can staff it

Five gates with a named accountable person at each, plus a named deputy for each, is between five and ten appointments. **An institution too small to make them will operate a compressed form of the model**, and this specification does not say which compressions preserve its properties and which destroy them. The one compression that certainly destroys it is holding G2 and G3 by the same person, which 2.3 forbids for that reason.

8.3 The central claim is untested and the comparison is not obvious

The model asserts that a named accountable person at a checkpoint changes the deployment outcome relative to an equivalent automated check. **That has not been measured**, and the alternative is not a straw man: a well-built pipeline with the same entry criteria enforced automatically is a serious competitor for the same job, and may be faster and more consistent. 8.6 states what would settle it.

8.4 No readiness instrument and no templates

There is no scored assessment against this model, so an institution can be told what the gates require and cannot be positioned against them. There are no published gate evidence templates, so what a G3 sign-off looks like is left to each institution to invent, and two institutions operating Five-Gate will produce records that are not comparable.

Both are known gaps rather than oversights, and both are the highest-value additions available to a future version.

8.5 The name has not been cleared

A trademark clearance search has not been completed, and the naming rule at 2.6 is stated as a discipline rather than as the outcome of one. This specification does not assert that the mark is available, and 2.6 records why it names no proprietor of the adjacent third-party mark.

8.6 What would falsify the model

The model is falsified if institutions operating five gates with a single named accountable person at each are found to deploy unfit AI systems at the same rate, and with the same severity of consequence, as institutions operating an automated delivery pipeline enforcing equivalent entry criteria with no named approver.

That is the comparison that matters, and it is deliberately the hardest one rather than a comparison against no process at all. If a named person adds nothing over an automated check on the same criteria, the gates are latency and the model should be replaced by the pipeline.

Three narrower findings would falsify parts of it. If systems whose G2 and G3 are held by one person fail no more often than those where the roles are separated, 2.3's separation requirement is ceremony. If institutions that patch after an incident recover as well as those that reopen validation, 4.2 is cost without benefit. And if gate records are found in practice not to join with validation and residency records during a reconstruction, the BOE mapping at Section 5 does not deliver what it claims.

8.7 Research questions

1. Does a named accountable person at a checkpoint change the deployment outcome relative to an equivalent automated check on the same criteria?
2. What does producing the five gate records cost, and at what institutional scale does it stop being proportionate to the risk?
3. Which compressions of the model preserve its properties in an institution too small to staff it fully?
4. Do institutions that reopen validation after an incident experience fewer repeat incidents than those that patch?

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A reader able to answer any of these from real deployment evidence is invited to make contact through the ORCID record in the front matter.

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9. Versioning

The major number changes when a change could make a previously conformant deployment non-conformant: a gate is added, removed or reordered, an entry criterion is added, or a requirement in Section 6 is tightened.

The minor number changes for additions and clarifications that cannot, including a new supplier named for an existing criterion, a corrected limitation, or the addition of a readiness instrument or evidence templates.

The gate identifiers G1 to G5 are normative and MUST NOT be renumbered. A gate retired in a future version retains its identifier, which is marked retired rather than reused.

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10. References

Every reference was verified against a live primary source on 29 August 2026, each with a deliberate failing control, because a check that cannot fail proves nothing.

[1] Khan, N. A. (2026). *The MESA Framework: A Four-Altitude Diagnostic Model for Institutional AI Governance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109836

[2] Khan, N. A. (2026). *A Pattern Language for Production LLM Platforms: Governed Routing, Agent Orchestration, and AI-Native Delivery. Pattern Language Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109864

[3] Khan, N. A. (2026). *The Defensible AI Framework Registry: Canonical Names, Definitions and Relationships for the Governed Production AI Discipline. Registry Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170112

[4] Khan, N. A. (2026). *AI Governance and Compliance Frameworks for the Middle East: The Enterprise Playbook.* First Edition. iSystematic Inc. ISBN 978-1-0678960-1-0

[5] International Organization for Standardization and International Electrotechnical Commission (2023). *ISO/IEC 42001:2023, Information technology. Artificial intelligence. Management system.* First edition, December 2023.

[6] National Institute of Standards and Technology (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* NIST AI 100-1, 26 January 2023.

[7] Office of the Superintendent of Financial Institutions, Canada (2025). *Guideline E-23, Model Risk Management (2027).* Published 11 September 2025, effective 1 May 2027. Scope includes artificial intelligence and machine learning models and models or data sourced from third parties.

[8] Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency and Federal Deposit Insurance Corporation (2026). *SR 26-2, Revised Guidance on Model Risk Management.* Issued 17 April 2026. Supersedes and replaces SR 11-7 (2011) and SR 21-8 (2021).

One verification could not be completed. No trademark register was reachable in a way that discriminated a genuine record from a deliberately invented control, so no ownership claim regarding any third-party mark appears in this document. Section 2.6 records that and explains why the distinction it draws does not depend on one.

Appendix A. Publication Provenance

A.1 First public version of this specification. Not peer reviewed. Deposited on Zenodo, which confers a DOI, a timestamp and citability. **Zenodo does not confer standard status, and nothing in this document uses the word standard for itself.**

A.2 The model was named and used before this specification, principally in [4] and across the author's practice. This document is its first standalone, versioned statement.

A.3 No preprint identifier exists and no preprint relation is asserted in this record's metadata.

A.4 Publication order. This record is third of eight and publishes before the Defensible AI Framework Registry [3], which publishes last because it cites the final identifier of every record it registers. Registry entry REG-06 remains the canonical statement of this model's relationships and governs where the two describe the same relationship. The registry's concept DOI is reserved in advance and is carried by reference [3]; it resolves when that record is published.

A.5 The author operates a consulting practice applying this model, which is a conflict of interest stated here. The mitigations are the same lesser ones available to any single-author instrument: the gate identifiers cannot be renumbered, the falsification conditions are published, and the limitations at Section 8 are stated rather than inferred. **None of these makes the specification independent.**

Document History

Version	Date	Summary
1.0	30 August 2026	First public version. Five gates with entry criteria, accountable roles, exit criteria, rejection paths and evidence records; two reverse paths; the BOE mapping; self-declared conformance; the naming and scope paragraph at 2.6

About the Author

Nabeel Khan is an enterprise AI architect and governance advisor, and Founder & Solutions Architect at iSystematic Inc. He has spent twenty-five years architecting enterprise data and machine-learning systems in government, healthcare, financial services and telecommunications across three continents, grounded in TOGAF, DMBOK, ISO 27001 and SOC 2.

He works on governed production AI; systems whose authority is bounded by the architecture, whose decisions carry the evidence needed to defend them, and whose behaviour the institution can explain afterwards. The frameworks specified here, and the regulatory knowledge behind them, are published as machine-accessible AI expertise and are callable over MCP without a key at mcp.nabeelkhan.com/api/mcp, so a reader may query this specification's contents directly and receive answers carrying their source.

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(Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22170122
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The DOI given is the concept DOI and always resolves to the latest version. Cite it unless you need to pin one specific version.

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